

Modular Pi Cam

This is a 3D printed camera based on the Raspberry Pi ecosystem. This camera features a Pi Zero 2W and a V3 camera module (28mm equivalent). This camera has an 18650 battery inside it with a battery life of around 6-7 hours depending on usage. This camera can take pictures and record video with sound (if a USB mic is plugged in). The camera software is written in python and uses systemd to run that code on boot.

For more information see

<https://github.com/jdc-cunningham/modular-pi-cam/tree/master/cameras/pi-zero/large-display>

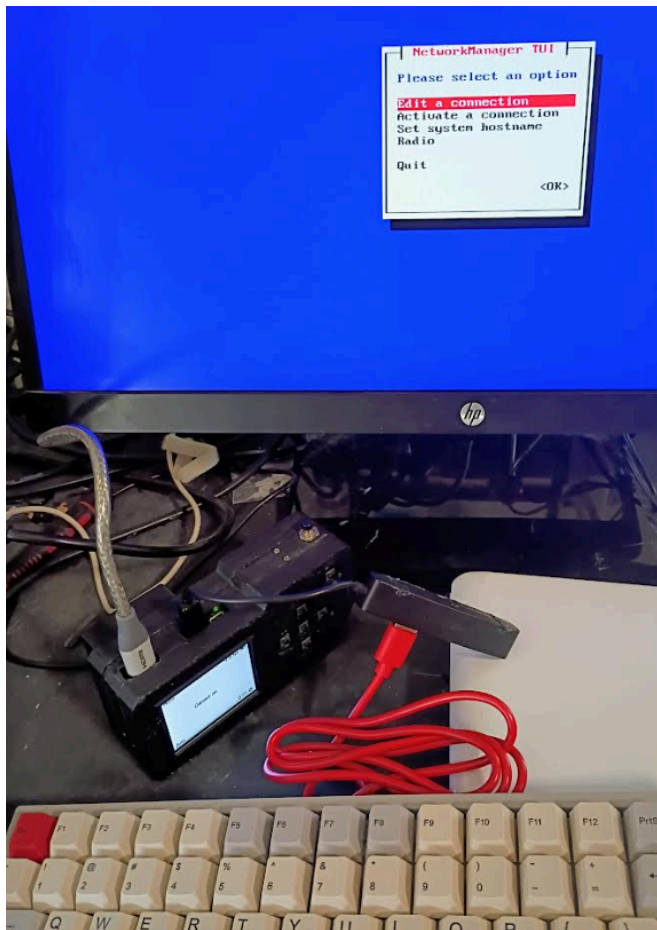
Login

username: pi

password: pi

WiFi

If you have the camera connected to a monitor/keyboard and are logged in. You can connect to your own WiFi using **\$sudo nmtui** (note \$ is not included in the command). Then you go through the dialogue to pick and activate your WiFi SSID.



Operation

If you're plugging in a USB device, make sure the camera is off first, plug the USB in, then turn the camera on. There is a power-draw issue when a USB device is plugged in and it won't work correctly if a USB device is plugged into the camera while it's running. A keyboard is fine. Turn the camera on and wait. Since this is a full computer vs. an embedded system, it has a heavier operating system so it takes a lot longer to start eg. around 20-30 seconds. Once the camera is on (you see the home menu) then you can start taking pictures by tapping the shutter button to engage the camera preview. Tap the shutter again to take a picture.



The V3 camera module has digital focus control (use the up/down buttons on the D-pad). F0 means everything is in focus and the higher the F number is the closer you have to have an object to focus it (also means background gets blurrier). Toggle mode (top left) to switch to video mode (click center button D-pad). You can view files on the camera with the files option. It's very basic, allowing you to see most recent photo and if you click to the right on the d-pad it'll show you older photos.

Lenses

If you do decide to get an HQ cam, you'll need to either buy a CCTV lens for this camera (C-mount) or old C-mount lenses like vintage cine lenses (may need flange adjustment). You can also get adapters for full frame lenses eg. Fotasy Nikon F to C-mount adapter.

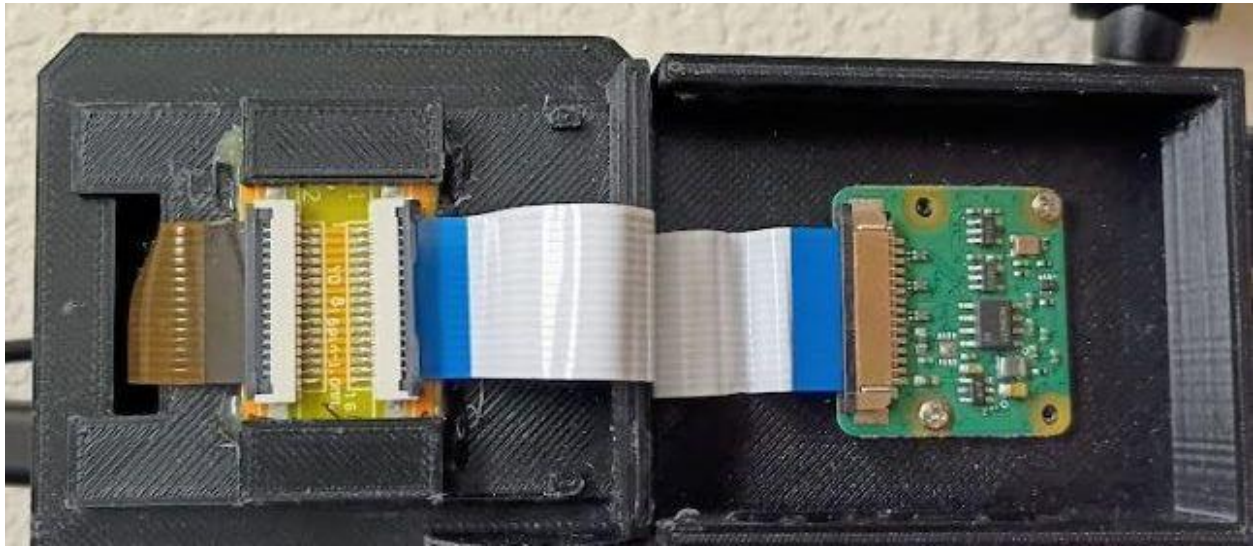
Charging

You need a micro usb cable and then a phone charger or PC USB port. The charging port will turn red when it's charging and blue when done.

Using an HQ camera

This camera is called “modular” since it can change cameras. In the past the back plate could be changed as well. It is important to get the cable oriented correctly or else you risk shorting the pi/camera. See the image below. This is a special cable where the connections are not flipped (blue tabs are on same side). This cable and the 3D printed mount for a Raspberry Pi HQ camera are included (along with 4 screws). The software will auto-detect what camera module is being used.

Slide the mount upwards to detach it, it snaps in place.



Getting files off the camera

Included is an OTG cable where you can plugin a USB (must be FAT32 formatted) and other peripherals like a keyboard and mic. The USB microphone (included) is used for video recordings. If you need to modify the camera’s software you can by plugging the camera to a monitor using a mini-HDMI cable to a full-sized HDMI and a keyboard. Note the USB position middle port. The one on the right is for powering the pi which is not required since it runs on battery. In the settings there is an option to transfer files to USB.

